

2025 United Kingdom Collision Dashboard

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1. Summary

The project was centered around creating an interactive dashboard for 2025 UK collision data. The data comprises about 45 columns, spanning from January 2025 to December 2025. The main objective was uncovering temporal, long term, environmental and geographical patterns. To achieve these objectives various initiatives were taken. Firstly, the data was decoded using its guide to facilitate understanding and provide more context. Necessary data procedures were used to prepare the data for analysis and visualization, some of the cleaning steps included dropping columns with more 40% missing data. Dropping columns with more than 40% missing data ensured non bias visualization. Missing data from other non-numeric columns were filled with “unknown”. After data cleaning procedures, initial exploratory data analysis (EDA) and principal component analysis (PCA) were carried out. On the dashboard to uncover geographical patterns maps were used, to uncover short term patterns hourly plots were used and to uncover environmental patterns weather related plots were used. In summary, key features on the dashboard were filters, maps, trend analysis and severity insights. Even though some deployment challenges were faced, the dashboard was successfully deployed. The dashboard could be very useful to public safety researchers, policymakers, commuters, and transport analysts.

2. Background research

Before implementation, various dashboards and visualization tools were researched. Criteria for the research were road safety, traffic analytics and public datasets dashboards. In cases where the criteria were met, strengths and weaknesses were noted. This gave a rough idea of what to do and what not to do.

2.1 FHWA Roadway safety data dashboard

The first dashboard analyzed was the FHWA Roadway safety data dashboard. This dashboard used maps and charts to facilitate the exploration of crashes, fatality and injury datasets. The dashboard was made by the U.S government. Some of the strengths of this dashboard include;

- A very large dataset was used, comprising of about 15 years of collision data
- The presence of custom filters based on vehicle type, location etc.
- Dashboard could be used for research and policy analysis.

Despite all its strengths the dashboard had some weaknesses, some of the weaknesses included;

- The dashboard was not beginner friendly making it very hard to use.
- Dashboard was focused just on the U.S.

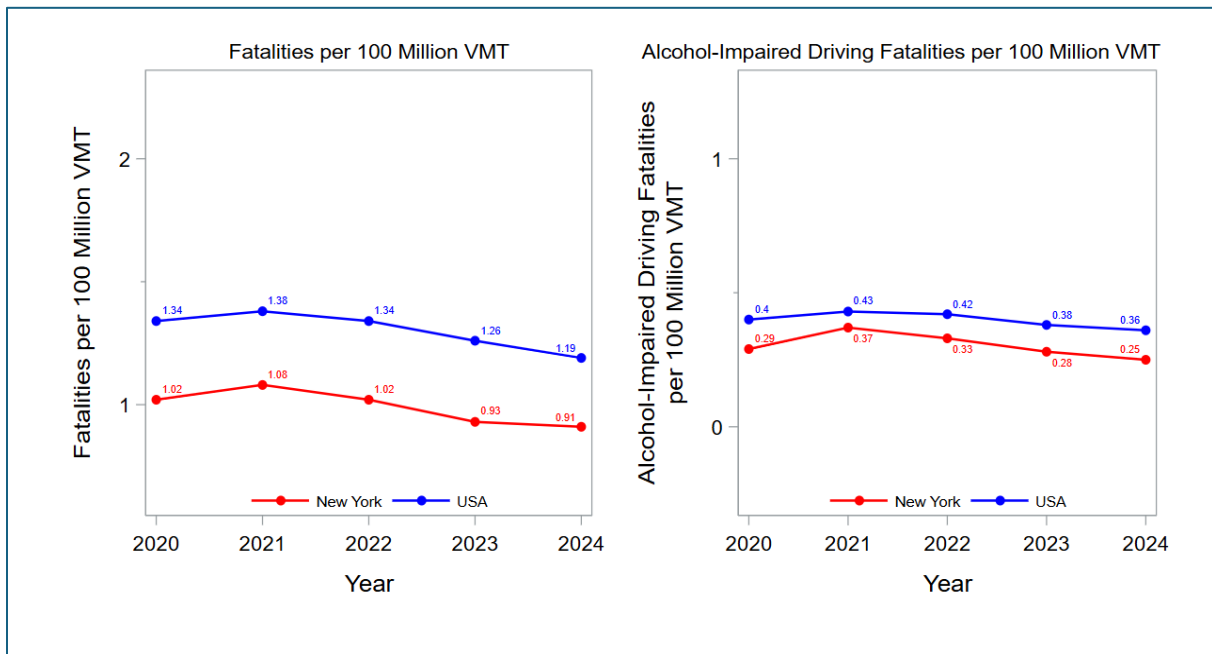


Figure 1(sample FHWA Roadway safety data dashboard)

This dashboard highlighted the importance of simplicity in modern dashboards as it was difficult to use, and it also demonstrated how filters can be used efficiently. These conclusions motivated the use of multiple filters and the creation of a straightforward dashboard.

2.2 Transport for London (TFL) dashboard

Furthermore, the Transport for London (TFL) road danger reduction dashboard was analyzed. The dashboard explored collision locations and road danger disparities in London. The dashboard included the following strengths;

- Very urban centric, could be useful for urban planning.
- Dashboard is very interactive and user friendly.

- Dashboard analyzes safety, enforcement and inequality insights.
- Good usage of maps.

The main limitation is its geographically limited to London. Its limited geographical location highlighted the need for a dashboard that covers a larger geographical location. The dashboard inspired the combination of various metrics rather than focusing solely on collision.

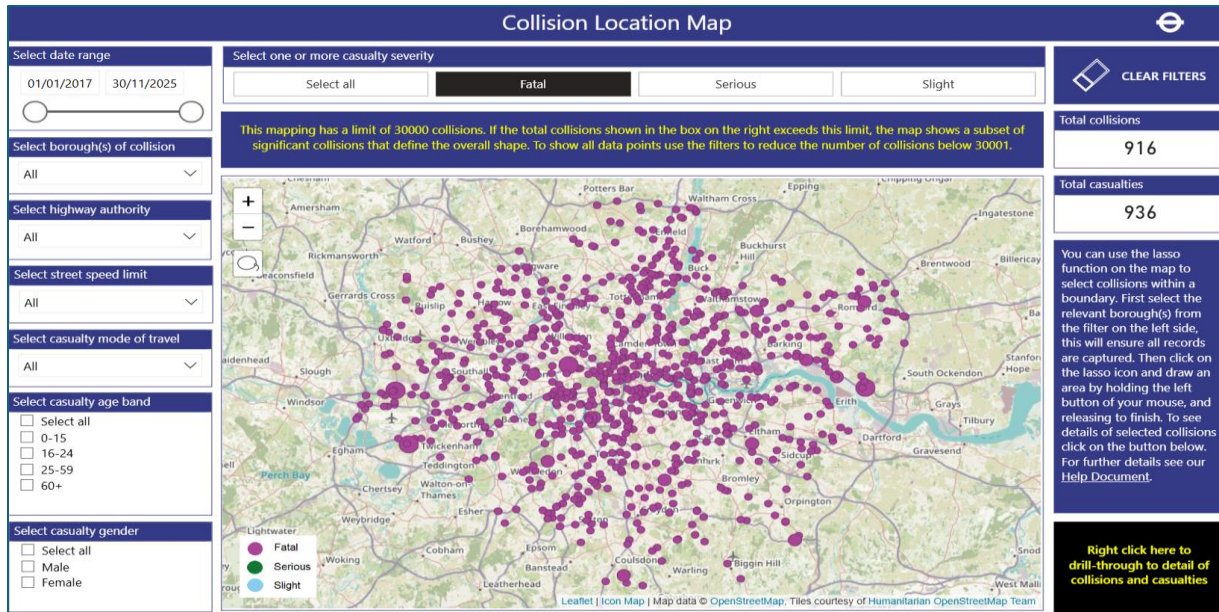


Figure 2 (sample TFL dashboard)

3. Data source and description

The data used was the UK collision data spanning from January 2025 to December 2025, the data contains about 45 columns and 48473 rows. This dataset was put together by the UK government .Each row in the dataset represents a collision and is independent of other rows, some of the columns used for analysis were; Longitude, Latitude, collision severity, number of vehicles, date, day of the week, time, road type, speed limit, weather condition and road surface condition. Collision severity was one of the most used columns, this column is a categorical column that represents how serious the collision was (fatal, serious and slight). While weather conditions represent the state of the weather during the collision, and road surface conditions represent the state of the road surface during collision. This database was deemed suitable due to its diverse structure of categorical and continuous columns, the dataset also covered a larger geographical area, unlike the TFL road danger reduction dashboard.

4. Data cleaning and preparation

Even though the collision dataset seemed very suitable for visualization, all its data was encoded making it unsuitable for visualization unless decoded. Numeric columns were converted to numeric data formats, and the data was then decoded using the guide file. After decoding the data columns with more than 40% missing data were dropped. Missing data from categorical columns were filled with “unknown” to maintain authenticity. Missing data from location related columns were filled with the mode, this was done because there was not a lot of missing data in location related columns.

The date column was then converted to date format, and the month and hour columns were then engineered. Other columns engineered were the high risk and wet dark columns, both columns were Boolean columns. If collision

severity was fatal or serious high risk would be true, otherwise it would be false. The wet dark column would be true if the road surface was wet or damp and the light condition was dark otherwise it would be false. These columns were engineered to provide more insight. Finally, some exploratory data analysis and some principal component analyses were carried out.

5. Visualization approach and Implementation

When implementing visuals a data frame was rendered to give the user an idea of the data they will be visualizing. Maps were then used to display crash locations and crash details. Clustered maps were utilized to display collision clustered or concentrations. Bar charts compared different groups or categories while line charts tracked trends, fluctuations and changes over time. Heatmaps were used to spot patterns and relationships.

To implement the dashboard the following tools were used: Python, Shiny for Python, Plotly and ipyleaflet. On the dashboard's introduction page, a brief description of the project and the data source is given, this page also displays 1000 rows from the dataset. The collision severity maps page displays severity by location while the clustered collision map displays collision clusters. However, all maps do not display the full dataset, due to UI complexity related to displaying maps. The primary plots page provides more plots for patterns recognition and categorical distribution. In the advanced filters page monthly and hourly line plots are provided alongside multiple filters for extensive exploration. Finally, the other plots page provides supplementary plots to the dashboard.

6. Evaluation and Criticism

6.1 Key challenges

Implementing the dashboard came with so many challenges, some of which:

- Decoding the datasets.
- Handling data inconsistencies during cleaning process.
- Performance issues related to map rendering and plots.
- Deployment challenges, mainly related to the dataset's path and requirements.txt.

6.2 Strengths

- The dashboard is very user friendly.
- Dashboards are very interactive.
- The dataset is real and from a credible source.
- Good visual story telling.
- The use of various visual types.
- Cleaning techniques did not affect data authenticity.

6.3 Weaknesses

- Maps did not display the full dataset, introducing the potential for bias. This could be solved in the future by using libraries that can support large datasets.
- Dashboard focused solely on the UK. This could be tackled by utilizing a collision dataset that focuses on various geographical locations.

6.4 Technical limitations

Maps faced various complications when trying to display the full dataset. That is why all maps showcased approximately 1000 collisions. The map that struggled the most was the clustered map. Even with 1000 collisions, both maps still struggled. All other plots seemed very slow due to the large dataset.

7. Deployment

A requirements.txt was created, this file included all packages the app will need when running. Then the dataset was then uploaded to a GitHub repository, and the raw GitHub link was used as the file path. A posit extension was installed visual studio code. Using the posit extension the app was then deployed.

The process might seem straightforward but multiple drawbacks were encountered. There were issues with the requirements.txt file and it had to be continuously updated to fix errors. During early deployment stages the app could not find the data source even though it was in the same folder as the app, so the data source was later deployed to GitHub and the raw link used as the data source.

Link to Dashboard: [Content Home - Posit Connect Cloud](#)

8. Version History

-  Version 1: Implemented a broken cluster map.
-  Version 2: Fixed cluster map and implemented severity map.
-  Version 3: Added legend to severity map and implemented the primary plots page.
-  Version 4: Implemented the advanced filter page.
-  Version 5 implemented other plots page.
-  Version 6: Created the introduction page.
-  Version 7: Styled the entire dashboard.
-  Version 8: Gave proper titles and labels to all plots.
-  Version 9: Deployed dashboard.

9. Conclusion

Even though drawbacks were faced the project achieved all its objectives by uncovering temporal, long term, environmental and geographical patterns. A highly interactive user-friendly collision dashboard was created. To enhance comprehension, a diverse range of plots were used. Dashboard accessibility was highlighted as an important quality as it helps users better understand the dashboard. The dashboard was deployed to ensure accessibility throughout the world. The deployment process provided insights into dealing with data source and configuration (requirements.txt) issues. In the future different map libraries will be used to display full data and a bigger collision data set will be used, one that covers the UK and other areas around the world. Even though the dataset is UK centric, some patterns can be studied and used for policy making in other areas similar to the UK. Ultimately, the dashboard can be very insightful to drivers, commuters, policy makers, transport analysts, and public safety researchers.